



Investigation report

Haul truck and light vehicle collision, Bindarra Gold Mine - AAIB-style causal and contributory factors report

Drafted 08/10/2026

INVESTIGATION WORKFLOW SUITE

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Synopsis

At 02:47:12 on 12/08/2026, HT-407, a 220 t class rigid haul truck, was descending empty on the main ramp of the Stage 4 open pit at Bindarra Gold Mine. A light vehicle, LV-118, drove out of the RL 312 dewatering spur and turned right across its path. The operator braked and steered towards the wall, slowing from 37 km/h to 24 km/h before the truck's front left struck the right side of LV-118. LV-118 was pushed 18.4 m down the ramp, rolling onto its left side against the inner windrow. Its driver was a survey subcontractor's grade control technician on his third night shift. He sustained a fractured left clavicle, three fractured left ribs and a small pneumothorax. He was extricated through the rear 27 minutes later and flown to the metropolitan trauma centre. The light vehicle was destroyed, but its roll-over protection structure was not breached and his seatbelt was fastened. The truck, damaged at the front left, returned to service on 14/08/2026. The Stage 4 pit stood down for 9 hours 15 minutes and the main ramp below RL 340 was closed for 22 hours.

Neither driver was very likely aware of the other until the truck came over the crest, about six seconds before impact. The 41 m sight line, against a 96 m stopping distance, very likely contributed. The spur had been cut under a works order, which excluded it from management of change, so nothing was referred. This very likely contributed. The intersection therefore had no stop control or warning sign despite failing the plan's sight distance rule, which likely contributed. The traffic management plan required positive calls only at listed intersections and allowed additions only at the annual review. It very likely made the intersection more dangerous. Neither driver made a call, which likely contributed.

A geofence extended up the ramp in June 2026, unknown to the operators, suppressed the collision awareness system at the spur. This likely contributed through the operator's expectation that a quiet system meant an empty ramp, rather than through the alert itself. The light vehicle driver slowed and looked, applying a give-way rule, but did not stop. This very likely contributed. His access card had been issued without a check of the pit driving assessment, which likely contributed. The external guideline left the mine to define when its plan should be reviewed and which intersections required a call. It likely contributed by making inaction the default, though it did not prevent the mine acting.

The haul roster of fourteen consecutive night shifts increased risk. So did the night shift extrication arrangement, which depends on a second responder called from the village. Neither was found to have contributed. Whether airborne dust delayed the operator's sighting of LV-118 could not be determined. Whether the operator was fatigued could equally have been so or not.

From 05:10 that day, the Mining Manager required a positive call at every Stage 4 intersection. The spur was closed.

Factual information

History of the occurrence

The RL 312 dewatering spur is a 240 m access road in the Stage 4 pit at Bindarra Gold Mine, cut under works order WO-26-4471 to reach dewatering bore DW-14. The works order recorded the earthworks complete on 31/07/2026. On 02/08/2026 the bore was energised and light vehicles first used the spur. The same day, the "ROAD CLOSED - CONSTRUCTION" sign at the spur mouth was removed and nothing replaced it.

At about 22:10 on 04/08/2026, a dewatering technician drove out of the spur in front of a descending haul truck. He reversed back into the spur, let the truck pass, and reported the event that night.

On 09/08/2026 the light vehicle driver, a grade control technician employed by a survey subcontractor, was issued a site access card. He began the first of three night shifts that evening.

On the night of 11/08/2026, haul truck HT-407 was hauling from the Stage 4 loading face up the main ramp to the ROM pad. Dispatch called the pit back to work after crib at 02:22:41 on 12/08/2026. HT-407 completed one cycle, and at 02:46:21 its operator called the three-sixty crossover, "empty, going down through".

The light vehicle driver left the DW-14 collar in LV-118 at about 02:45 and reached the spur mouth at about 02:47. Nobody on the shift knew he was in the pit. In his account, he slowed at the mouth, looked up the ramp, saw nothing and turned right to go up the ramp. He did not believe he came to a complete stop. The turn took LV-118 across the descending lane with its right side towards a truck coming down. Neither driver made a radio call before entering the intersection.

HT-407 came over the crest above the intersection at 37 km/h. At 02:47:06 the truck was 53 m above the spur mouth, the first position from which the survey found the light vehicle's beacon to be in view. The operator applied the service brakes, took the retarder to maximum and steered right towards the wall. At 02:47:12 the truck's front left struck the right side of LV-118 at 24 km/h, 6.2 m into the ramp from the spur mouth. The truck pushed the light vehicle 18.4 m down the ramp. At 11.2 m it rolled over, coming to rest on its left side against the inner windrow, facing down-ramp. The truck stopped 19 m below the intersection at 02:47:19.

The operator reported the collision at 02:47:41. At 02:48:01 dispatch stopped all Stage 4 units and told him to stay in his cab. Released at 02:48:44, he went to the light vehicle driver, who was conscious and trapped with his door against the ground. The emergency response team was paged at 02:48 and the first response vehicle arrived at 02:54. The second responder arrived from the accommodation village at 03:07 with the hydraulic rescue tool. The driver was extricated through the rear of the vehicle at 03:14, 27 minutes after the collision. He reached the surface medical facility at 03:41 and was flown to the metropolitan trauma centre at 07:40.

The driver sustained a fractured left clavicle, three fractured left ribs and a small pneumothorax. The Stage 4 pit stood down for 9 hours 15 minutes, and the main ramp below RL 340 was closed for 22 hours. The occurrence was notified to the State Mines Inspectorate as a reportable injury.

Recorded information

The contractor's technical services group downloaded the fleet management system and collision awareness system data from HT-407 at 09:30 and provided them to the investigation unopened. The pit channel recording, RAD-2026-0332, agreed with the fleet management system to within 0.4 s. All times are local.

The fleet management system download, TRW-TS-2026-114, recorded:

- 02:47:06.0: the last steady data point, at 37 km/h at RL 317, 53 m above the spur mouth.
- 02:47:07.4: 38.5 m above the mouth, service brake pressure rose from 0 to 620 kPa in 0.4 s, the retarder went to maximum and 14 degrees of right steering was applied.
- 02:47:09.2: 32 km/h, 21.4 m above the mouth.
- 02:47:12.0: the collision, at 24 km/h.

The operator's response time was 1.4 seconds.

The collision awareness system event log recorded:

- "18:47:09.200 PROXIMITY HT-407 tag LV-118 range 21.4 m closing 32 km/h zone S4-LOAD alert SUPPRESSED (geofence)" [collision awareness system event log, UTC]

Nothing was presented in the cab. Outside a suppression geofence, the system would have raised an advisory alert when the tag entered the 60 m detection ring, at about 02:47:06.5 on the download.

The pit channel recording captured the operator's first call, "I've hit a light vehicle on the ramp at three-twelve", and the supervisor's instruction at 02:49:38 not to move the driver and to wait for the emergency response team.

Equipment information

HT-407 is a 220 t class rigid haul truck and was travelling empty. It recorded 0.80 m/s² in the monthly brake performance test on 03/08/2026, and its mean deceleration from brake application to impact was 0.78 m/s² net of the grade. The retarder and service brakes performed as specified. Damage was confined to the front left bumper, the left mudguard and the lower two access ladder steps. The truck returned to service on 14/08/2026.

Its collision awareness system operated at Level 8 (operator advisory). Level 9 (automated intervention) is available on the fleet but is not enabled at Bindarra. The in-cab fleet management system map for the Stage 4 pit, last published on 24/07/2026, did not show the RL 312 spur.

LV-118 is not dispatched, carries no fleet management system, and has a passive collision awareness tag. It was destroyed. Its roll-over protection structure was deformed inward at the right rear mounting but not breached. The driver's seatbelt was found fastened, with loading marks on the webbing. The 4.5 m amber beacon whip was sheared at its base, and the lamp worked when tested after recovery.

Personnel information

The haul truck operator had 7 years on 220 t class trucks and 14 months at Bindarra. His verification of competency was current from 06/2025, with an annual refresher in 02/2026. He had received no traffic management refresher since his site induction, whose module names the nine listed intersections and does not cover unlisted ones. No person on site held training on the collision awareness system's suppression geofences. He was on night 10 of a 14 night swing, 8 hours 47 minutes into a 12 hour 30 minute shift, and had slept 6 hours 30 minutes the preceding day. He described himself as "Tired, the way you are on night ten. Not falling asleep tired."

The light vehicle driver had 6 years in grade control. He completed his contractor's two hour online general mining induction on 24/07/2026, and his site access card was issued after a 20 minute gatehouse briefing. There is no record that he completed the Bindarra site-specific supplement or the light vehicle pit driving assessment. He stated that he had not heard of the

assessment, had not seen the traffic management plan, and had not been told to make a radio call before entering the ramp.

The night shift supervisor had 19 years in open pit mining, 2 years 6 months in the role, and was on the same roster as his crews.

The haul truck operator and the supervisor both returned negative drug and alcohol tests.

Site information

At RL 312 the main ramp descends at 9.1 per cent, is 24 m wide, and was dry and well maintained. The spur joins on a descending truck's left, on the inside of a left-hand bend, at an included angle of 71 degrees, 34 m below a convex grade break that crews call the crest.

From the haul truck operator's eye height of 5.80 m, the spur mouth comes into view at 41 m and the top of the light vehicle's beacon at 53 m. From the spur mouth, a driver first sees a descending truck at 38 m. For the last 25 m of the spur, the 1.6 m inner windrow blocks the view of the ramp. The survey recorded that "The intersection does not meet clause 6.2 on either limb", which requires a sight distance of not less than 120 m. The truck's calculated stopping distance at 40 km/h was 96 m.

The intersection had no warning sign, no give-way or stop control, no delineation and no posted speed for the spur.

The drivers' accounts of visibility differed. The haul truck operator said some dust was hanging but he could not say whether it made a difference. The light vehicle driver said the air in the spur was clear.

Organisational and management information

The Bindarra traffic management plan, BIN-TMP-004, requires positive radio calls at nine intersections listed at Appendix B. Clause 6.4 requires signage and give-way control only at intersections on the plan's road layout drawing. The RL 312 spur was on neither.

Annual reviews were completed in 11/2023, 11/2024 and 11/2025. Clause 9.1 provides for an out-of-cycle review, but none has been carried out in the plan's eleven year history. Since the 12/11/2025 review, 14 new roads had been cut in the Stage 4 pit. Eleven were cut under a works order, and none of those 11 was referred to the plan's owner.

Works order WO-26-4471 classified the spur as "Management of change not required, ORR-PRO-022 clause 5.3", and its handover field read "N/A".

The collision awareness system's suppression geofence, S4-LOAD, was drawn on 14/03/2025 around the loading face. On 09/06/2026 it was extended to a boundary 180 m up-ramp of the intersection, as a configuration task not reviewed against the traffic management plan. The fleet's standing alert report does not include suppressions.

Contractor Management Standard ORR-STD-009, clauses 7.3 to 7.5, requires the pit driving assessment to "be completed at site by every person who will drive a vehicle in an operating pit", and requires the engaging department to verify completion before a card is issued. The gatehouse issued cards against the recognition agreement alone.

No permit, call-in or other process let the supervisor of an operating pit know which light vehicles were in it.

Other information

Fictional case study for training. Not a real investigation.

Hazard report HAZ-2026-0917 described the near miss of 04/08/2026, stating that "you cannot see up the ramp from that road" and that "I do not think they know the road is there". On 06/08/2026 the Dewatering Superintendent closed it after a crew reminder and reduced its risk rating from Medium to Low. It was not referred outside the dewatering department.

Three earlier reports, from 03/2025, 09/2025 and 02/2026, each described a light vehicle meeting a haul truck unexpectedly at an unlisted intersection. Each was closed the same way, and none led to a change to the plan or to any physical control.

At 05:10 on 12/08/2026 the Mining Manager issued a site-wide direction requiring a positive call before entering any intersection in the Stage 4 pit, listed or not.

Analysis

External Influences

Guideline sets no review trigger and no basis for the positive-call list

The Bindarra traffic management plan was written within an external frame set by the State Mines Inspectorate. That frame is Mines Safety Guideline MSG-14, revision 4, made under regulation 6.22 of the Mining Operations Safety Regulations 2018. Compliance with the guideline is one way of discharging the regulation, which itself requires only that "the operator shall manage the risks arising from the interaction of vehicles at the mine". The text of both is documentary and not in dispute. It is virtually certain that the guideline stood in the form described below on the night of the occurrence.

Two clauses bear on this matter. Clause 3.2 requires the plan to be "reviewed at intervals not exceeding twelve months, and following any significant change to the mine's road network, vehicle fleet or operating pattern". The guideline does not define a significant change, and it gives no example of one. Clause 5.1 requires the operator to "nominate the intersections, crossings and other interaction zones at the mine at which positive communication between vehicles is required before entry". It sets no minimum and gives no basis on which the nomination is to be made. Clause 3.5 adds that the plan "is a site document" which is neither approved by nor submitted to the Inspectorate except on request after an occurrence or an inspection. Both the review trigger and the basis for the positive-call list were therefore left to the mine, and nobody outside the mine examined either. Clause 5.4 offers five alternative means of making a light vehicle detectable, any one of which satisfies it. It is context for this matter rather than part of it.

On its face, the Bindarra plan met the guideline. It had been reviewed on 12/11/2025, within twelve months, and Appendix B nominated nine interaction zones. The mine was not in breach of the guideline. Nothing in this analysis suggests a failure by the Inspectorate. The point is the shape of the discretion the guideline leaves, and how that discretion was taken up at site.

The plan filled that discretion with a closed loop. Under clause 4.3.1, a positive call was required only before entering "any intersection listed at Appendix B". An intersection reached Appendix B only where it appeared on the road layout drawing at Schedule 1, and both documents were reissued only at the annual review. No provision allowed an intersection to be added between reviews. No provision required a new road to be referred to the plan's owner. The RL 312 spur was not on the Schedule 1 drawing dated 12/11/2025.

The plan's history shows that this loop was not limited to one road. Annual reviews were completed in 2023, 2024 and 2025. No out-of-cycle review had been held in the plan's eleven years. Fourteen new roads had been cut in the Stage 4 pit since the last review. The three cut under a management of change record were referred to the plan's owner, and none of the eleven cut under a works order was. Over the 24 months before the occurrence, the reporting system returned four hazard reports describing a light vehicle and a haul truck meeting unexpectedly at intersections not listed at Appendix B. Each was closed within three days with a reminder to the crew. None reached the plan's owner, and none led to a change to Appendix B, to Schedule 1 or to any physical control.

The case for influence runs as follows. Suppose the guideline had defined a significant change and set a basis for choosing positive-call intersections. A new intersection on a haul road would then have had a defined route onto the call list between annual reviews. An intersection failing the plan's own sight distance rule would also have had a claim on that list. The eleven unREFERRED roads would have had somewhere to land. As written, the guideline left both terms to the same site document, reviewed once a year, with no outside eye on either.

The case against influence is also real. The guideline did not stop anybody from acting. Clause 9.1 of the mine's own plan allowed an out-of-cycle review at any time, and MSG-14 did not prevent the mine from listing every intersection it had. On this reading, the mine did not fail to act because the guideline was vague. Nothing inside the mine ever put the question in front of the plan's owner, so the external clause is background and the internal arrangements are the whole of the matter.

Weighing the two, the investigation found it likely that the guideline influenced the occurrence. That conclusion sits at the threshold of the standard of proof and is not settled beyond it. It supports a narrow proposition. Because the guideline left the trigger and the basis to the site document, it made inaction the default, even though the mine could have acted on its own.

The guideline's influence runs along one path only. It bears on the arrangement under which positive calls were required only at listed intersections, and intersections were added only at the annual review. That arrangement is argued under Risk Controls. The guideline has no bearing on the opening of the spur under a works order. Whether a works order was exempt from management of change was set by the mine's own procedure and not by MSG-14. That matter is argued under Organisational Influences.

Accordingly, the absence from MSG-14 of a defined review trigger and of any basis for nominating positive-call intersections virtually certainly existed and likely contributed, through the listing path alone, to the RL 312 intersection remaining off the positive-call list, which places it as a causal factor held at the standard of proof rather than beyond it.

Organisational Influences

Spur opened under a works order, so nothing was referred to anyone

The procedure, the works order and the road register agree. It is virtually certain that the RL 312 spur was opened without a management of change record.

ORR-PRO-022 clause 5.2 requires such a record before "a change to the mine's road network". Clause 5.3 excludes "routine earthworks carried out under a works order within an approved pit design, where no new plant or vehicle type is introduced." WO-26-4471 relied on that exclusion. It described the 240 m access to bore DW-14 as "Routine earthworks within the approved Stage 4 design" and recorded "Management of change not required, ORR-PRO-022 clause 5.3." Its handover field reads "N/A".

The consequence lay in what the exclusion removed. Clause 5.4 places three referrals inside the management of change record:

- the traffic management plan owner
- the survey department, for the fleet management system road network
- the contractor's technical services group, for the collision awareness system configuration.

The works order form at Schedule 4 has no equivalent fields, and none of the three referrals was made. The in-cab map published on 24/07/2026 was still current on the night and showed no intersection at RL 312. The S4-LOAD suppression geofence had been carried up the ramp on 09/06/2026, and it already covered the point where the spur was later joined to it.

Had a record been raised, the referral to the plan's owner would have come with a survey showing that the intersection failed the clause 6.2 sight distance rule. It would therefore have needed a stop control and the clause 6.4 signs. The referral to the technical services group would have reached the people who configure the geofence the intersection was cut inside. The road register shows that these referrals work in practice. Of the fourteen new roads cut in Stage 4 since the 12/11/2025

review, the three raised under a management of change record were referred to the plan's owner. None of the eleven raised under a works order was.

There is a limit to this argument. A referral is not itself a control, and the plan's owner might have left the spur to the annual review. No out-of-cycle review has been held in the plan's eleven years. A referral also could not have changed the rule confining positive calls to listed intersections, which is considered with the risk controls.

It is very likely that opening the spur under a works order led both to the absence of a stop control and signs at the intersection and to the suppression of collision alerts there, and it was a causal factor.

Access cards issued with no check of the pit driving assessment

It is virtually certain that the light vehicle driver was given a site access card without any check that he had completed the Bindarra light vehicle pit driving assessment.

The standard itself is sound. ORR-STD-009 clause 7.4 states that equivalence "does not extend to the Bindarra site-specific supplement, or to the Bindarra light vehicle pit driving assessment, which shall be completed at site by every person who will drive a vehicle in an operating pit and shall not be substituted." Clause 7.5 makes the engaging department responsible for verifying this before a card is issued. What failed was the place where that verification had to happen. The gatehouse issues cards against the recognition agreement alone and is not asked for either item. Nothing in the card system distinguishes a person who may drive in an operating pit from one who may not.

The driver's card was issued on 09/08/2026 after a 20 minute briefing. His training file shows the Hallorn online module of 24/07/2026 and no pit driving assessment. He had not heard of the assessment and had not seen BIN-TMP-004. [NOT ESTABLISHED: whether the Bindarra site-specific supplement was issued to the light vehicle driver.]

The case for influence rests on what the assessment adds to what he actually did. It teaches a driver at a mouth he cannot see out of to stop and look again, rather than to slow. On the recorded times, a stop at the mouth would very likely have shown him the truck.

The case against is substantial. He was already applying the assessment's first rule, that light vehicles give way to heavy vehicles at all times, and he slowed and looked before he went. The plan requires no call at an unlisted intersection, so the assessment would not have taught him to call at RL 312. He said he would have called had he been told to, but that cannot be extended to an intersection where no call was required. The content of the assessment was not available to the investigation, so the mechanism is inferred from its stated purpose and not from its text.

On balance it is likely that the absence of the check led to the light vehicle driving out of the spur without stopping. It was a causal factor, although the mechanism is inferred and the finding is held below the others for that reason.

Haul operators rostered for fourteen consecutive 12.5 hour night shifts

It is virtually certain that the Stage 4 haul roster was 14 nights on and 7 off, worked from 18:00 to 06:30 local. Each shift lasted 12 hours 30 minutes, with a 30 minute crib.

The operator was on night 10 of 14. The collision at 02:47 came 8 hours 47 minutes into the shift, inside the window of lowest circadian alertness. Every limit in the contractor's fatigue standard was met, and none of the previous nine shifts ran beyond 12 hours 40 minutes.

The argument that the roster played a part is that it placed the operator at the wheel at the point where degraded detection and slower response are expected. Picking out a hazard in an unlit pit is the kind of task that fatigue degrades first.

The recorded data answer that argument. The operator responded in 1.4 seconds, inside the 1.5 seconds the plan's own stopping distance calculation assumes, and the braking and steering that followed were correct and immediate. Braking at the recorded 0.78 m/s² from 53 m, even a response time of zero would still have ended in a collision, at about 17 km/h instead of 24 km/h. The response cost about 7 km/h of impact speed, not the collision itself. The download shows no decision or omission through which the roster could have acted. It is therefore unlikely that the roster played any part in this occurrence. Whether the operator was fatigued on the night is a separate question, which remains unresolved and is discussed under Local Conditions.

The roster still matters as a standing exposure. On every swing it places each operator of a 220 tonne vehicle at the wheel in the small hours of a tenth, twelfth and fourteenth consecutive night. At an unlisted intersection, the defence against an unlit hazard is the operator's own detection and response. Where the sight line is longer than at RL 312, a slower response could be the difference between stopping and not stopping.

The roster was likely a safety issue that increased risk to haul truck operations, independently of this collision.

Risk Controls

Collision alerts suppressed at the spur by a geofence operators knew nothing of

The collision awareness system on HT-407 was fitted and working. At 02:47:09.2 it logged the light vehicle at 21.4 m, closing at 32 km/h, and suppressed the alert because the RL 312 intersection lay inside the S4-LOAD geofence. The configuration, the event log and the fleet statistics come from the system itself, so the suppression is not in doubt. That operators did not know of it rests on the accounts of the truck operator and the night shift supervisor, and on the absence of any training record or notice. Taken together, it is extremely likely that alerts were suppressed at the spur without the operators' knowledge.

The alert itself would have achieved little. Unsuppressed, it would have sounded at about 02:47:06.5. That was half a second after the beacon came into view and about a second before the brakes were applied. It would therefore have told the operator what he was already reacting to, and the system is advisory only. What remains is his expectation. From 09/06/2026 the system was silent on the ramp whatever was on the road, and nobody told him. He wrote in his own report that he "took that to mean the pit was empty when it was quiet". His in-cab display showed no intersection, and there was none marked on the road. The silence removed one more reason to think anyone could be below the crest.

Against this, nothing shows that the expectation changed what he did. He was descending at the speed every operator used on that ramp, and the remedy he named was a sign, not the system. Both readings stand, because the effect is real but cannot be measured. The suppression is placed as a causal factor, acting through expectation rather than through the alert. Its origin in the opening of the spur is argued under Organisational Influences.

It was likely that the suppressed alerts influenced the occurrence through the operator's expectation that a silent system meant an empty ramp.

Positive calls required only at listed intersections, added only at the annual review

BIN-TMP-004 requires a positive call only at the nine intersections listed at Appendix B. An intersection reaches that list only through the drawing at Schedule 1, and both documents are reissued only at the annual review. Nothing requires a new road to be referred to the plan's owner. The RL 312 spur was on neither document, and the earliest date it could have been listed was 12/11/2026. These matters are documentary, and the gap is virtually certain to have existed.

The pit channel recording shows the practice the plan produced. HT-407 called, and was acknowledged, at the RL 360 crossover going up and coming down and at the ROM pad turn-off. The last of these calls was "407, three-sixty crossover, empty, going down through." at 02:46:21, 51 seconds before the collision. Neither vehicle made any call about the spur. The truck operator, the light vehicle driver and the night shift supervisor each independently described calling at the nine intersections on the crib room board and nowhere else. Had RL 312 been listed, the operator would have called there, on a channel the light vehicle was using. The only reservation is that a call made and not acknowledged still leaves a driver deciding whether to go. This is a causal factor. Why the plan carries no review trigger is argued under External Influences.

It was very likely that the absence of any requirement to call at the spur influenced the occurrence.

No stop control or warning sign at an intersection failing the sight distance rule

Clause 6.2 of BIN-TMP-004 applies to every intersection, listed or not. The survey gave a sight distance of 41 m against a stopping distance of 96 m, so the clause required a stop control, signalised control or traffic controller. The intersection had none of these. It also had no warning sign, no give-way control, no delineation and no posted speed. The construction sign was removed on 02/08/2026 and nothing replaced it. HAZ-2026-0917 had named the missing sign and give-way seven days earlier. The absence of control is virtually certain.

On the recorded times, the truck was within the light vehicle driver's 38 m of view from about 02:47:07.4. That was nearly two seconds before he reached the mouth. A full stop there would very likely have shown him the truck, and he could have let it pass, as the technician did on 04/08/2026. The operator said, "If there'd been a sign on my side of that crest I'd have been at twenty and I'd have been looking for him."

The case against is real. A distance of 38 m is about four seconds of a truck at 37 km/h, which is a thin margin even from a standstill. The driver was already slowing and looking under the give-way rule he described. The effect of a sign rests on the operator's own view after the event. A sign at an intersection absent from the in-cab map would also have been a single cue. For these reasons the influence is held at likely and no higher. The factor is causal, acting on the light vehicle's entry to the ramp, which is argued with the individual actions.

It was likely that the absence of a stop control and warning sign influenced the light vehicle's entry onto the ramp.

Night shift extrication waits on a second responder called from the village

The night shift is rostered with one mines rescue paramedic. The second responder must be present before any extrication, and is called in from the village 11 km from the gatehouse. That responder brings the vehicle carrying the hydraulic rescue tool. On the night, the second responder arrived at 03:07 and the driver was extricated at 03:14, 27 minutes after the collision. The post-incident review measured the delay at approximately 9 minutes from 02:58, when the primary survey was complete. The arrangement is virtually certain to have existed.

The collision and the injuries were complete before the team was paged. The review recorded on clinical grounds that the delay did not affect the outcome, so it is unlikely to have influenced this occurrence. The arrangement still matters. The driver was conscious, stable, restrained and protected by an intact roll-over structure. With a fuel fire, an unconscious casualty or a compromised airway, the same nine minutes could produce a different outcome, and nothing in the arrangement scales with severity.

It was very likely that the reliance on a second responder from the village increased risk, and it is reported as a safety issue that played no part in this occurrence.

Local Conditions

Sight line of 41 m at the spur against a 96 m stopping distance

The mine survey department surveyed the intersection on 13/08/2026 at the eye heights of both vehicles. The survey placed the spur mouth 34 m below the convex grade break that the crews call the crest. From the truck's cab, at an eye height of 5.80 m, the top of the light vehicle's 4.5 m beacon whip first came into view at 53 m. The spur mouth came into view at 41 m, and the vehicle's body was hidden until then. From the light vehicle, at an eye height of 1.45 m, a descending truck first came into view at 38 m, and the 1.6 m inner windrow hid its lower body for the last 25 m of the spur's approach. The survey concluded: "The available sight distance at this intersection is 41 m. The intersection does not meet clause 6.2 on either limb."

The hazard report raised seven days earlier described the same view independently. The technician wrote that "you are out on the road before you can see anything". It is virtually certain that the sight line was as surveyed.

The comparison that matters is with the truck's stopping distance. A 220 t class truck empty at the posted 40 km/h on this grade needs 96 m to stop. The plan's 120 m minimum is a matter of compliance and adds nothing to that physical comparison. On the recorded data, the truck needed about 68 m on brakes alone from 37 km/h at the net deceleration of 0.78 m/s² it achieved. With twice the sight line, the same 1.4 second response and the same braking would have stopped the truck short of the spur. At 41 m, a truck at the posted speed could not stop for a vehicle at the intersection, however alert its operator.

A sight line becomes dangerous only once something is in it. The absence of a stop control and warning signs, which put the light vehicle in the truck's path, is argued under Risk Controls. It is very likely that the sight line contributed to neither driver being aware of the other until the truck came over the crest. The evidence places the 41 m sight line against a 96 m stopping distance as a causal factor in the collision.

Airborne dust on the ramp above the intersection during the truck's approach

Three pieces of evidence support the presence of dust:

- The water cart last watered that section of ramp 1 hour 37 minutes before the occurrence. The standing interval is 90 minutes, so a pass was 7 minutes overdue.
- The fixed dust monitor on the RL 328 bench logged PM10 at 142 micrograms per cubic metre in the ten minutes covering the occurrence, against a shift mean of 88. The monitor is 180 m up-ramp of the intersection and above the crest.
- The truck operator described dust hanging on the descent. He was driving behind his own lights into an unlit pit.

Against this, the reading was well below the site trigger of 250. It measured particulate at the monitor head, not what could be seen from either cab, and nothing at Bindarra measures visibility. The light vehicle driver was firm that the air was clear. His view, however, was of the spur below the windrow, not of the ramp above the crest where the truck was. On that basis it is likely that airborne dust was present on the ramp above the intersection.

Whether the dust delayed the operator's sighting of the beacon could not be settled. Dust in a truck's lights over a crest is the condition in which a driver picks out a beacon late. The case against a delay is nonetheless strong. The download puts the truck at its last steady data point 53 m above the spur mouth, which is where the survey says the beacon could first be seen at all, assuming clear air. A sighting at the geometric limit leaves dust nothing to explain. The operator himself declined

to say that the dust made a difference. His 1.4 second response from that point was inside the plan's own 1.5 second allowance.

What prevents the matter being excluded is that the sighting point was inferred from the data and the survey rather than recorded. The evidence therefore cannot rule out dust costing a fraction of a second at the edge of the lights. Settling the question would need a measurement of what could be seen from the cab on the night, and no such measurement exists. The question of whether dust affected the sighting remains unresolved, and nothing from it is carried into the conclusions.

Ramp surface wet or slippery at and above the intersection

A wet or slippery ramp surface was considered and found very unlikely: the met station recorded no precipitation and no rain for 26 days before, the investigator found the surface dry throughout at 06:20, and the braked tyre scuff ran continuously for 58 m at a deceleration matching the unit's tested dry figure.

Haul truck operator fatigued during the approach to the intersection

The operator's duty pointed towards fatigue. He was on night 10 of a 14 night swing and 8 hours 47 minutes into a 12 hour 30 minute shift. The time was 02:47, within the window of lowest circadian alertness. His own sleep record showed between 5 and 7 hours on each of the preceding days and 6 hours 30 minutes on the last. He described himself as tired in the way of a tenth night. Detecting a beacon in an unlit pit is a task that fatigue degrades early.

Against this, the operator described the tiredness as ordinary and said he was awake and driving properly. Every limit in the contractor's fatigue standard was met. His recorded performance showed no sign of impairment: a 1.4 second response and immediate, correct braking and steering. Being tired on a tenth night is not the same as being fatigued to a degree that degrades performance. Apart from his own account and the recorded response, no evidence bears on the question either way. It is about as likely as not that the operator was fatigued, and the matter remains unresolved.

A detailed fatigue assessment would have been needed to resolve it. That assessment would not have changed the outcome of the analysis. Braking at 0.78 m/s² from 53 m with no response time at all still ends in a collision, at about 17 km/h rather than 24 km/h. The risk carried by the roster itself is considered under Organisational Influences. Nothing from this question is carried into the conclusions.

Individual Actions

Neither driver made a positive call before entering the intersection

It was virtually certain that neither driver made a positive call before entering the RL 312 intersection. The pit channel was logged continuously, and the investigator listened to the whole period from the return from crib to two minutes after the collision. There was no transmission from either vehicle about the spur, the dewatering bore, or a light vehicle being anywhere in the Stage 4 pit. LV-118 made no transmission at all that shift after logging on at 18:04. All three interviews confirmed the same.

The omission has to be read against what the truck did make. HT-407 called at every listed intersection it passed, and each call was acknowledged. Its last was at 02:46:21, 51 seconds before the collision: "407, three-sixty crossover, empty, going down through." The operator was therefore using the channel exactly as the plan required. BIN-TMP-004 clause 4.3.2 says of an intersection not listed at Appendix B that "No call is required", and the RL 312 spur was not listed. The truck operator, the light vehicle driver and the night shift supervisor each described the same practice

independently: calls are made at the nine intersections on the crib room board and nowhere else. The supervisor stated that he had never corrected anybody for not calling elsewhere. The omission therefore had no individual owner. It was the practice the document produced, and the reasons the spur was absent from Appendix B are argued under Risk Controls, where the listing arrangement was found to be a safety issue.

A call from either vehicle would have been heard by the other, by dispatch and by the supervisor. The dispatcher was already tracking the truck closely enough to place it. The argument is limited in one respect, however. A call gives each driver knowledge of the other, not separation, and the two vehicles would still have had to settle who went first over 41 m of sight line. For that reason the influence is put at likely and not higher.

The absence of a positive call likely contributed to neither driver being aware of the other until the truck came over the crest, and it is treated as a causal factor at that level of confidence.

Light vehicle drove out of the spur onto the ramp without stopping

It was virtually certain that the light vehicle left the spur without stopping. The driver first described stopping at the mouth. On read-back he corrected that to slowing. He looked right up the ramp, saw nothing, and drove out, turning right to go back up the ramp. He put his speed out of the spur at about 25 km/h and did not believe he had come to a complete stop. That correction ran against his own interest, and his account agreed with the collision awareness log and the scene measurements. The log placed LV-118 at the spur mouth at 02:47:09.2, with the truck 21.4 m above it. The impact followed 2.8 seconds later, 6.2 m into the ramp from the mouth.

What the driver could see from where he looked explains why he saw nothing. From the mouth, at his eye height, the survey gives 38 m of sight line to a descending truck. Before the mouth, the inner windrow hid the truck's lower body for the last 25 m of the spur. On the recorded times his look came a second or two before the truck came over the crest, when it was beyond anything his position allowed him to see. The truck was within his 38 m from about 02:47:07.4, nearly two seconds before he reached the mouth. He was applying the rule in clause 4.1, that light vehicles give way to heavy vehicles at all times, and he looked for one. Nothing at the mouth told him the look was insufficient. There was no stop or give-way control, no stop line, no warning sign and no posted speed. He had not completed a Bindarra pit driving assessment and had never seen the traffic management plan. Those conditions are argued under Organisational Influences and Risk Controls, where the issue of access cards without the assessment and the absence of a stop control were each found to be safety issues.

Without the light vehicle on the ramp there would have been no collision, and driving out without stopping was the proximate step. Had he stopped at the mouth and looked again, the truck was already inside his view, and he would very likely have seen it and let it pass. The dewatering technician did exactly that on 04/08 by reversing back into the spur. The only reservation is that a stop is only as good as the look taken from it, and that look had 38 m to work with.

It was extremely likely that the light vehicle driving onto the ramp without stopping contributed to the collision, and it is a causal factor.

Truck operator braked and steered right within 1.4 seconds of the beacon appearing

It was virtually certain that the truck operator responded within 1.4 seconds. The download recorded the last steady data point at 02:47:06.0, at 37 km/h and 53 m above the spur mouth, which is where the survey places the first sight of the beacon. From 02:47:07.4, at 38.5 m, service brake pressure rose from 0 to 620 kPa in 0.4 seconds. The retarder went to maximum and 14 degrees of right steering was applied in the same movement. That response was inside the 1.5

seconds the plan's own stopping distance calculation allows. Mean deceleration to the collision was 0.78 m/s² net, against the unit's tested 0.80. The braked scuff at the scene deviated 1.4 m right over its first 20 m, which corroborated the steering input.

The response did not prevent the collision, and on a 41 m sight line it could not have. It did reduce the impact speed from 37 km/h to 24 km/h, which left about 0.42 of the impact energy. Steering right turned the truck away from the open edge, where the pit edge and a drop of more than 100 m lay to its left. The push therefore ran down the ramp, and the light vehicle came to rest against the pit side windrow and did not go through it. After stopping 19 m below the intersection, the operator reported the collision, stayed in the cab as dispatch instructed until the ramp was stopped, and then remained with the driver until the response team arrived.

The operator's braking and steering very likely reduced the severity of the outcome, and they are the principal reason the collision produced a survivable injury.

Occurrence Events

Neither driver aware of the other until the truck came over the crest

Every line of enquiry in this investigation arrives at one condition. Two vehicles were converging on the RL 312 intersection at night, and each was unknown to the other until about six seconds remained. The conditions that produced this are argued elsewhere in this analysis. They are the sight line at the spur, the suppression of collision alerts by the S4-LOAD geofence, the absence of positive calls, and the light vehicle entering the ramp without stopping. They are treated here as a single mechanism, because that is how they acted.

The haul truck operator had no means of knowing that the spur existed. The in-cab road network had last been published on 24/07/2026, a week before the spur was cut, and it showed no intersection at RL 312. There was no warning sign on the approach and no call on the channel. The collision awareness system raised nothing. Its log shows that it detected the light vehicle and suppressed the alert:

- "18:47:09.200 PROXIMITY HT-407 tag LV-118 range 21.4 m closing 32 km/h zone S4-LOAD alert SUPPRESSED (geofence)" [Collision awareness system event log, HT-407]

The light vehicle driver was approaching a mouth from which, at his eye height of 1.45 m, the survey found "first has line of sight to a descending truck at 38 m". There was no give-way control and no call, and he had no knowledge of the truck. His look up the ramp came before the truck was within his view.

The first sighting available to the truck operator was the light vehicle's beacon over the crest. The survey placed that sighting at 53 m: "The top of a 4.5 m beacon whip on a light vehicle approaching the mouth in the spur is in view from the same eye height from 53 m". The download recorded the truck's last steady data point at that position at 02:47:06 local (18:47:06 UTC). The collision followed six seconds later. The radio log, the download, the collision awareness log and all three interviews were examined. None of them gave any indication that either driver knew of the other before that moment. The night shift supervisor did not know that a light vehicle was in the pit, and there was no permit, call-in or board by which he could have known.

The mutual unawareness was extremely likely to have existed. The one reservation is that the light vehicle's movements before it reached the spur mouth rest on the driver's own account alone. It was extremely likely that the mutual unawareness led to the collision. Once the two vehicles were converging with six seconds left, nothing between that point and the impact could have prevented it. The evidence therefore puts the mutual unawareness of the two drivers as a causal factor in the collision.

The haul truck's front left struck the right side of the light vehicle at 24 km/h

The download recorded a deceleration spike at the spur intersection at 02:47:12 local (18:47:12 UTC), at 24 km/h. This was 4.6 seconds after the brakes were applied. The truck's accelerometer logged 1.9 g longitudinal. The scene examination placed the impact 6.2 m into the ramp from the spur mouth, 3.1 m out from the toe of the inner (pit side) windrow.

LV-118 was turning right out of the spur to go up the ramp. That turn took it across the descending lane with its right side towards the truck. The damage agreed with this. Damage to LV-118 was concentrated on the right side from the B pillar forward. Damage to HT-407 was confined to the front left bumper, the left mudguard and the lower two steps of the access ladder, with nothing above the bumper line. Right-side glass and mirror fragments lay from 4.8 m to 14 m down-ramp of the impact position.

The intersection geometry made a side impact inevitable for a vehicle making that turn. A side impact from a 220 tonne vehicle on a light vehicle leaves the outcome to the light vehicle's structure, the restraint and the speed. The operator's braking and steering, discussed under Individual Actions, reduced the impact speed from the 37 km/h that would otherwise have applied. The recorded data, the damage pattern and the debris field all agree. It was virtually certain that the impact occurred as described, and virtually certain that it was the mechanism by which the convergence became an injury. The impact was a causal factor in the outcome.

The light vehicle was pushed 18 m down the ramp and rolled onto its left side

A continuous gouge ran 18.4 m down-ramp from the impact position. At 11.2 m it changed from tyre scuff to body panel gouge, and that is where the vehicle went over. LV-118 came to rest on its left side against the inner windrow, facing down-ramp. The driver sustained a fractured left clavicle, three fractured left ribs and a small pneumothorax, and was conscious throughout. The door on the resting side was against the ground, so extrication was through the rear of the vehicle.

Two defences worked. The driver's seatbelt was found fastened and the webbing showed loading marks. The roll-over protection structure was deformed inward at the right rear mounting but was not breached. The injuries were those that a side impact and a roll-over produce when the structure holds and the restraint is worn. Had either been absent, the outcome would very likely have been different. The scene measurements, the vehicle examination and the medical record agree. It was virtually certain that the light vehicle was pushed and rolled in this way, and virtually certain that the roll-over produced the injuries that made this a serious occurrence. It was a causal factor in those injuries.

Service brakes delivered less than their tested deceleration during the stop

It was very unlikely that the brakes underperformed and the hypothesis was ruled out, because the 0.78 m/s² achieved during the occurrence lay within the scatter of the 0.80 and 0.82 m/s² measured in brake tests on 03/08/2026 and 14/08/2026, and the unit carried no brake, retarder or park brake defect.

Conclusions

The investigation identified the following causal and contributory factors. They should not be read as apportioning blame or liability to any organisation or individual.

A finding stated without a likelihood term was established as fact.

Findings

1. Mines Safety Guideline MSG-14 required the traffic management plan to be reviewed after any significant change to the road network without defining such a change. It also left the operator to nominate its own positive-call intersections, with no minimum and no basis for the nomination. Bindarra's plan met the guideline on its face, and the mine was not in breach of it. (Safety issue)
2. The RL 312 spur was cut under works order WO-26-4471, which classified the work as routine earthworks excluded from management of change by ORR-PRO-022 clause 5.3. As a result, none of the three referrals required at clause 5.4 was made. These referrals were to the plan's owner, to the survey department for the in-cab road network, and to the contractor's technical services group for the collision awareness system geofences. (Safety issue)
3. Fourteen new roads had been cut in Stage 4 since the last review of the traffic management plan. The three raised under a management of change record were referred to the plan's owner, and none of the eleven raised under a works order was. (Safety issue)
4. ORR-STD-009 clause 7.4 required every person who would drive in an operating pit to complete the Bindarra light vehicle pit driving assessment at site. The gatehouse issued access cards against the contractor's recognition agreement alone, and the access card system had no means of showing whether the assessment had been done. The light vehicle driver had not completed the assessment and had never seen BIN-TMP-004. (Safety issue)
5. The Stage 4 haul roster was 14 consecutive 12 hour 30 minute night shifts, and every limit in the contractor's fatigue management standard was met. The haul truck operator was on night 10 of 14 and was 8 hours 47 minutes into his shift at the time of the collision. (Safety issue)
6. On 09/06/2026 the S4-LOAD suppression geofence was extended up the main ramp to a boundary 180 m up-ramp of where the spur was later cut, and the extension was not reviewed against the traffic management plan. It is very likely that no operator or supervisor knew the collision awareness system was suppressed on the ramp, and no training on the suppression geofences was found for any person on site. (Safety issue)
7. The collision awareness system logged the light vehicle at 21.4 m at 02:47:09.2 and suppressed the alert. Unsuppressed, it would have alerted at about 02:47:06.5, about one second before the brakes were applied. The alert itself would therefore have told the operator little more than he was already reacting to.
8. BIN-TMP-004 required a positive call only at the nine intersections listed at Appendix B. An intersection could be added to the list only at the annual review, and no provision required a new road to be referred to the plan's owner. The RL 312 intersection was not listed, and the next date on which it could have been listed was 12/11/2026. (Safety issue)
9. The RL 312 intersection failed the sight distance requirement of BIN-TMP-004 clause 6.2, which in that case required a stop control, a signalised control or a traffic controller. It had no warning sign, no stop or give-way control, no delineation and no posted speed. The construction sign at the mouth was removed on 02/08/2026 and nothing replaced it. (Safety issue)

10. Hazard report HAZ-2026-0917 of 04/08/2026 described the sight line, the missing sign and the missing give-way at the spur. It was closed on 06/08/2026 with a reminder to the crew and was not referred outside the dewatering department. Three earlier reports describing light vehicles meeting haul trucks unexpectedly at unlisted intersections were closed in the same way. (Safety issue)

11. On night shift, extrication could not begin until a second responder arrived from the accommodation village 11 km away with the hydraulic rescue tool. The site's post-incident review measured the resulting delay at approximately 9 minutes and found that it did not affect the clinical outcome. (Safety issue)

12. The survey found the following first lines of sight:

- from a descending truck to the spur mouth, 41 m
- from a descending truck to the light vehicle's beacon, 53 m
- from the spur mouth to a descending truck, 38 m.

The stopping distance of the truck at the posted 40 km/h was 96 m.

13. The ramp surface was dry. No rain had fallen at the site for 26 days before the occurrence.

14. Neither driver made a positive call before entering the intersection, and none was required there. HT-407 had called at the listed RL 360 crossover 51 seconds before the collision.

15. The light vehicle driver slowed at the spur mouth, looked up the ramp, saw nothing and drove out at about 25 km/h without coming to a complete stop. He was applying the plan's rule that light vehicles give way to heavy vehicles. The truck was within his 38 m view of the ramp from about 02:47:07.4, nearly two seconds before he reached the mouth.

16. The haul truck operator applied the brakes and steered right 1.4 seconds after the beacon came into view, within the 1.5 seconds assumed in the plan's stopping distance calculation. His response reduced the impact speed from 37 km/h to 24 km/h.

17. It is very likely that neither driver was aware of the other until the truck came over the crest. The night shift supervisor did not know a light vehicle was in the pit, and no permit, call-in or other process existed by which he could have known.

18. At 02:47:12 the haul truck's front left struck the right side of LV-118 at 24 km/h, 6.2 m into the ramp from the spur mouth.

19. LV-118 was pushed 18.4 m down the ramp and rolled onto its left side. Its roll-over protection structure deformed without being breached, and the driver's seatbelt was fastened and showed loading marks. The driver sustained a fractured left clavicle, three fractured left ribs and a small pneumothorax.

20. The truck's service brakes and retarder performed to specification. They achieved 0.78 m/s^2 during the stop, against tested figures of 0.80 m/s^2 before the occurrence and 0.82 m/s^2 after it.

Causal factors

- With virtual certainty, the spur was opened under a works order, so nothing was referred to anyone. It is very likely that this contributed to the occurrence, because the referrals that would have reached the plan's owner, the in-cab road network and the geofence configuration were never made. (Safety issue)
- It is very likely that collision alerts were suppressed at the spur by a geofence operators knew nothing of. It is likely that this contributed, through the truck operator's expectation that a quiet system meant an empty ramp rather than through the alert itself, an effect that is real but cannot be measured. (Safety issue)

Fictional case study for training. Not a real investigation.

- The sight line of 41 m at the spur against a 96 m stopping distance existed with virtual certainty. It very likely contributed, because a truck at the posted speed could not stop for anything at that intersection.
- That the guideline sets no review trigger and no basis for the positive-call list is virtually certain. It is likely that this contributed by making inaction the default on the path by which an intersection reaches the call list. That assessment is held at likely and no higher, because the guideline did not prevent the mine from reviewing its plan or listing the intersection on its own initiative. (Safety issue)
- Positive calls were required only at listed intersections, added only at the annual review, and this is virtually certain. It very likely contributed, since the RL 312 intersection was not listed and no call was therefore required there. (Safety issue)
- It is virtually certain that neither driver made a positive call before entering the intersection. It is likely that this contributed, although a call would have given each driver knowledge of the other rather than separation between them.
- It is virtually certain that access cards were issued with no check of the pit driving assessment. It is likely that this contributed, with the qualification that the content of the assessment was not available to the investigation and the driver was already applying its first rule. (Safety issue)
- With virtual certainty, there was no stop control or warning sign at an intersection failing the sight distance rule. This likely contributed, although the 38 m view from the spur mouth left a thin margin even from a stop, and the effect a sign would have had on the truck rests on the operator's own account. (Safety issue)
- The light vehicle drove out of the spur onto the ramp without stopping, which is virtually certain. This very likely contributed, because the truck was already inside the driver's view from the mouth before he reached it.
- It is very likely that neither driver was aware of the other until the truck came over the crest, and very likely that this contributed, since the collision followed directly from that convergence with about six seconds remaining.
- The haul truck's front left struck the right side of the light vehicle at 24 km/h, which is virtually certain. It is virtually certain that this impact contributed to the injuries sustained.
- Virtually certain, too, is that the light vehicle was pushed 18 m down the ramp and rolled onto its left side. It is virtually certain that the roll-over contributed to the driver's injuries.

Contributory factors

- Haul operators were rostered for fourteen consecutive 12.5 hour night shifts, which is virtually certain. It is likely that this increased risk, as a standing exposure on every swing, even though the investigation found it unlikely that the roster contributed to this occurrence, given the operator's recorded 1.4 second response. (Safety issue)
- It is virtually certain that night shift extrication waits on a second responder called from the village. It is very likely that this increased risk, as a structural gap in the last line of defence on every night shift, although it is unlikely that the arrangement affected this occurrence or its outcome, the driver's injuries being stable and complete before the response was paged. (Safety issue)

Other findings

- It is very unlikely that the ramp surface was wet or slippery at and above the intersection, and this was examined and ruled out. No rain had fallen for 26 days, the surface was dry

when examined at 06:20, and the deceleration achieved matched the truck's dry tested figure.

- It is very unlikely that the service brakes delivered less than their tested deceleration during the stop, and this was examined and ruled out. The 0.78 m/s^2 achieved lay within the scatter of the 0.80 m/s^2 and 0.82 m/s^2 recorded in tests either side of the occurrence, and no brake defect had been raised against the unit in the preceding twelve months.
- The truck operator braked and steered right within 1.4 seconds of the beacon appearing, which is virtually certain. This very likely reduced risk, lowering the impact speed from 37 km/h to 24 km/h and keeping the light vehicle against the pit side windrow, away from an open edge with a drop of more than 100 m.

Two further matters were examined and could not be resolved. Neither is a finding.

It is likely that airborne dust was present on the ramp above the intersection during the truck's approach. Several things supported its presence:

- a water cart pass on that section was 7 minutes overdue
- the fixed dust monitor on the RL 328 bench logged PM10 at 142 micrograms per cubic metre against a shift mean of 88
- the operator described dust hanging on the descent.

The investigation could not determine whether the dust delayed the operator's sighting of the light vehicle. The reading was well below the site trigger of 250 and measured particulate at the monitor head, not what could be seen from either cab. The light vehicle driver described the air in the spur as clear. The download placed the truck at 53 m, the geometric limit of sight to the beacon, at its last steady data point. The operator's 1.4 second response from that point left no room for a delay, and he did not say that the dust made a difference. Against this, the moment of sighting was inferred from the data and the survey rather than recorded, so a delay of a fraction of a second at the edge of the truck's lights could not be excluded. Settling the question would require a recorded sighting point or a measure of visibility from the cab, and nothing at Bindarra measures visibility.

Whether the haul truck operator was fatigued to a degree that degraded his performance during the approach could equally have been so or not. Some evidence pointed towards fatigue:

- he was on night 10 of 14 and inside the window of lowest circadian alertness
- he had slept between 5 and 7 hours on each of the ten preceding days
- he described himself as tired.

Other evidence pointed away from it. He described the tiredness as ordinary, every limit in the fatigue standard was met, and his recorded response showed none of the usual signs of impairment. A detailed fatigue assessment would have been needed to settle the question, and none was carried out. Even had fatigue been established, braking at the recorded 0.78 m/s^2 from 53 m with no response time at all would still have ended in a collision, at about 17 km/h instead of 24 km/h.

Safety action and recommendations

Guideline sets no review trigger and no basis for the positive-call list

Mines Safety Guideline MSG-14 revision 4 required the traffic management plan to be reviewed after any significant change to the road network. It also required the operator to nominate the intersections at which a positive call was needed. It defined neither the change that should reopen the plan nor the basis on which an intersection should be nominated, and nobody outside the mine examined either choice. Bindarra's plan met the guideline on its face, and the mine was not in breach. The issue lies in the shape of the discretion the guideline leaves, not in any failure by the Inspectorate.

It is likely that this discretion contributed to the occurrence, and the finding is held at that level and no higher. A defined trigger would have required the plan to be reopened when a road was added. A stated basis for nomination would have given an intersection failing the sight distance rule a claim on the call list. Against that, the mine could have held an out-of-cycle review under its own clause 9.1, or listed every intersection, and the guideline prevented neither. Its effect was to make inaction the default.

That default matters beyond this spur. Eleven of the fourteen roads cut in Stage 4 since the last review were never referred to the plan's owner. No out-of-cycle review had been held in the plan's eleven years.

No safety action has been recorded against this issue. A site-wide direction requiring a positive call at every Stage 4 intersection was issued after the occurrence. It answers the mine's own call list rather than the guideline. No recommendation is made.

Spur opened under a works order, so nothing was referred to anyone

The mine's management of change procedure, ORR-PRO-022, requires a management of change record before any change to the mine's road network. Clause 5.3 excludes routine earthworks carried out under a works order within an approved pit design. Works order WO-26-4471 relied on that exclusion to cut 240 m of access to bore DW-14 at RL 312, which created a new intersection with the main ramp. No management of change record was raised, so none of the three referrals at clause 5.4 was made. Those referrals would have gone to the traffic management plan owner, to the survey department for the fleet management system road network, and to the contractor's technical services group for the collision awareness system geofences. The absence of those referrals very likely contributed to two outcomes: there was no stop control or warning sign at the intersection, and collision alerts remained suppressed there.

The issue extends beyond this spur. Fourteen new roads have been cut in Stage 4 since the 12/11/2025 review of the plan. The three raised under a management of change record were referred to the plan's owner, and none of the eleven raised under a works order was.

No safety action addressing this issue has been recorded. The site-wide direction issued after the occurrence requires a positive call at every Stage 4 intersection, but it does not change how a new road is referred.

Access cards issued with no check of the pit driving assessment

The site's Contractor Management Standard, ORR-STD-009, allows a recognised contractor's general induction to be accepted as equivalent to the site's own. It states, however, that this equivalence does not extend to the Bindarra light vehicle pit driving assessment. Every person who

will drive in an operating pit must complete that assessment at site, and the engaging department is to verify it before a card is issued.

In practice the gatehouse issued cards against the contractor's recognition agreement alone. It held no record of the assessment and was not asked for one. Nothing in the access card system distinguished a person who may drive in an operating pit from one who may not. It is virtually certain that the light vehicle driver was carded on that basis without having completed the assessment. The standard itself was sound. What failed was that the only point at which the verification could have happened had no provision for it.

The issue matters because the assessment is where a light vehicle driver is taught to approach an intersection he cannot see out of by stopping and looking again, rather than by slowing. It is likely that the absence of the check contributed to the light vehicle leaving the spur without stopping. The same gap applied to every contractor driver carded under a recognition agreement. Whether the driver was given the Bindarra site-specific supplement is [NOT ESTABLISHED: whether the site supplement was issued to the light vehicle driver].

No safety action has been recorded in response to this issue. The site-wide direction issued after the occurrence concerns positive calls at Stage 4 intersections and does not change how access cards are issued. No safety recommendation is made.

Positive calls required only at listed intersections, added only at the annual review

Under the traffic management plan BIN-TMP-004, a positive call on the pit channel was required only before entering one of the nine intersections listed at Appendix B. An intersection could reach that list only through the road layout drawing at Schedule 1, and both documents were reissued only at the annual review. Nothing required a new road to be referred to the plan's owner when it was cut. The RL 312 spur was on neither document. The earliest date on which it could have been listed was 12/11/2026, three months after the collision. This arrangement was a causal factor, and it very likely contributed to neither driver making a positive call before entering the intersection.

The control was the mine's own defence against exactly this interaction, and at this intersection its own terms left it unarmed. The gap was not confined to the spur. Eleven of the fourteen new Stage 4 roads cut since the last review had never been referred, and no out-of-cycle review had been held in the plan's eleven year history.

At 05:10 on 12/08/2026, two hours twenty-three minutes after the collision, the Mining Manager issued a site-wide direction. It required a positive call before entering any intersection in the Stage 4 pit, listed or not, until further notice. It answers the issue in part, because it leaves unchanged the route by which an intersection reaches Appendix B.

Collision alerts suppressed at the spur by a geofence operators knew nothing of

The collision awareness system on the Stage 4 haul fleet raised no alert at the RL 312 intersection because the intersection lay inside the S4-LOAD suppression geofence. The geofence boundary had been extended 180 m up the main ramp on 09/06/2026 as a system configuration task. Nobody outside the contractor's technical services group was told, and no training record on site covers the geofences. It is extremely likely that alerts were suppressed at the spur without the operators' knowledge, and likely that this contributed to the occurrence. The contribution lay in the operator's expectation that a quiet system meant an empty ramp. It did not lie in the alert itself, which would have sounded only about half a second after the beacon came into view. That effect

is real but cannot be measured. The issue extends beyond this occurrence: after 09/06/2026 the system suppressed nine proximity events involving a light vehicle on the main ramp below the boundary.

No stop control or warning sign at an intersection failing the sight distance rule

The RL 312 spur met the main ramp at an intersection with a surveyed sight distance of 41 m, against a stopping distance of 96 m. Clause 6.2 of BIN-TMP-004 therefore required a stop control, signalised control or traffic controller there. The intersection had none of these. Nor did it have a warning sign, a give-way control, delineation or a posted speed. The spur had been opened under a works order, so the intersection was never referred to the plan's owner. Referral was the route by which such a control would have been installed.

The absence of control was a causal factor. It is likely that a stop control at the spur mouth would have held the light vehicle there while the truck came into view, and that the driver would then have let the truck pass. The finding goes no higher because the margin from a stop at that mouth was thin, with only 38 m of view. The same absence had been reported in HAZ-2026-0917 seven days before the occurrence.

The operator, Orrabinda Resources, barricaded the RL 312 spur and closed it to all traffic on the night of the occurrence. This action answers the issue in full at the RL 312 spur.

No safety recommendation has been recorded for this issue.

Haul operators rostered for fourteen consecutive 12.5 hour night shifts

The contractor's Stage 4 haul roster was 14 nights on and 7 off. Each shift ran from 18:00 to 06:30 local, a period of 12 hours 30 minutes that included a 30 minute crib. The roster sat inside a fatigue management standard that permitted up to 14 consecutive night shifts with a minimum 10 hour break. The night shift supervisor worked the same pattern as the crews he supervised. On the night of the collision the haul truck operator was on night 10 of 14, and every limit in the standard had been met.

It was unlikely that the roster contributed to this collision. The operator's recorded response fell within the time that the plan's own stopping distance calculation assumes. The roster is nonetheless a standing exposure, built into the pattern of work rather than arising from one night, and it likely increased risk. On every swing it places each haul truck operator at the wheel of a 220 tonne vehicle in the small hours of a tenth, twelfth and fourteenth consecutive night. At an unlisted intersection, the defence against an unlit hazard rests on that operator's own detection and response. That is the kind of task that fatigue degrades first.

No safety action has been recorded in response to this safety issue. After the occurrence, a site-wide direction was issued requiring a positive call at every Stage 4 intersection. That direction concerns positive calls and does not address the roster. No safety recommendation is made in respect of this issue.

Night shift extrication waits on a second responder called from the village

The night shift at Bindarra is rostered with one qualified mines rescue paramedic, based at the surface medical facility. No extrication may begin until a second responder is present. That responder is called in from the accommodation village 11 km from the gatehouse, in the second response vehicle, which is garaged there and carries the hydraulic rescue tool. On the night of the

Fictional case study for training. Not a real investigation.

occurrence, the paramedic had completed the primary survey and made the extrication plan by 02:58. The second responder arrived with the tool at 03:07. The post-incident review measured the delay to the extrication at approximately 9 minutes.

The collision and the driver's injuries were complete before the emergency response was paged. The arrangement is therefore unlikely to have affected this occurrence or its outcome. The driver was conscious, stable, restrained and protected by an intact roll-over structure, and the review found on clinical grounds that the delay did not affect the outcome.

The arrangement is nonetheless very likely to be important to safety. It is a structural gap in a last line of defence, it is present on every night shift, and nothing in it changes with the severity of the casualty the team is called to. A fuel fire, an unconscious casualty, or an airway that needed the vehicle opened within the first ten minutes would each make the same wait matter.

No safety action has been recorded in response to this issue. [NOT ESTABLISHED: whether any safety action has been taken or is proposed, and whether a safety recommendation is to be made in respect of this issue]

Distribution

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